

Stellar parameters

**twinkle, twinkle little star
Oh, I know who you are!**

- PART I
 - What are the stellar parameters. Intro
 - Light is the great informant (spectroscopy)
 - Wien's law. Surface temperature and color
 - Chemical composition.
 - classification of stars



Sagittarius Star Cloud



HOW TO USE THE LIGHT TO LEARN ALL WHAT WE KNOW ABOUT THE STARS (STELLAR PARAMETERS) ? WHAT ARE THE STELLAR PARAMETERS ?

Stellar parameters are intrinsic properties of stars.

- **mass of stars**

Determine most of the stellar parameters except for chemical composition.

life of stars , size of stars , Luminosity , temperature = color

From binary stars system. Use Kepler's law (amended by Newton)

= relationship between:

The masses of the stars, the orbital period and the distance between the stars.

(more mass = more tug = smaller distance and larger speed).

We can use direct measurements or we can use the Doppler shift = relative radial speed

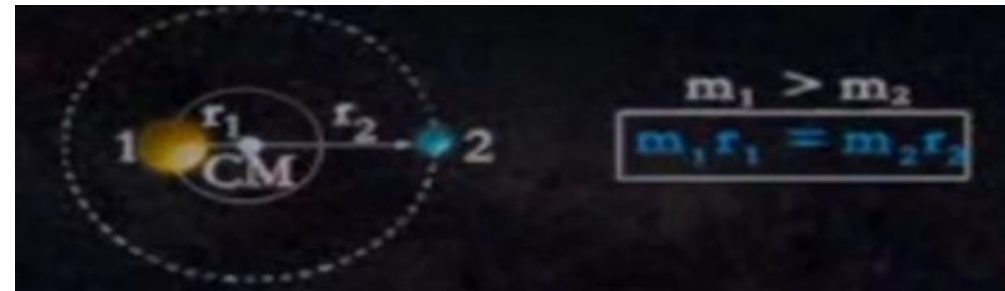
Kepler's law $M_1 + M_2 = a^3/p^2$,

$M_2/M_1 = V_1/V_2$ and $M_2/M_1 = r_1/r_2$

.a is the distance between the stars, p the orbital Period.

We can use Doppler shift or direct measurements.

Show video binary stars



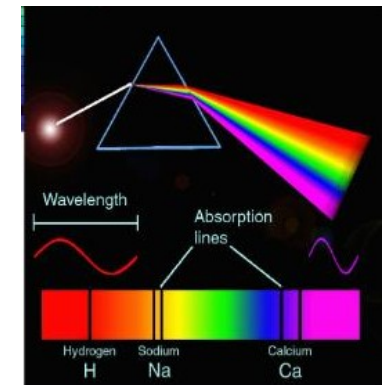
- **chemical composition** (also depends on mass but not only)

using the spectra of stars (fingerprints of atoms/molecules)

primarily determined by the elemental abundances

in the interstellar medium, which itself reflects t

he results of nucleosynthesis processes in earlier generations of stars.



- **color of a star /temperature** from peak wavelength of the **black body curve**.

$$T = 0.029 / \lambda \quad \text{Wien's law}$$

Example: Sun peaks in the green (about 10,000F),

The temperature of a star on the main sequence is **roughly proportional to its mass**

- **Luminosity black body curve** (area below the spectrum of a star)
(also called intrinsic brightness/absolute magnitude) → Power output of a star. →
Depends on the rate of fusion and size of stars.

Stefan-Boltzmann law. The energy per unit area and per unit second output
depends on the temperature to the 4th power. (question on test?)

Depends on mass.

$$L = 4 \pi R^2 \sigma T^4 \quad L = M^4$$

- **Life of stars.** It depends on mass. How fast its burning fuel.
fuel/(rate at which fuel is burnt). **life time (about) = $M/L = 1/M^3$**

Example: 10 times more mass than the Sun , its life is divided by 1000/
(Sun lives for 10 billion years. A 10 more massive star lives for only 10 million years)

Note: If $L \times 2$ then life is divided by 2 (about).

The light that burns twice as bright, burns Half as long.

Famous quote in blade runner (Ridley Scott, 1982, LA 2019, pianist future)

<https://www.youtube.com/watch?v=wRxHYHPzs7s>

- **Size of stars – radius** about proportional to mass for its main life (main sequence)

- **magnetic field and rotation rate.** (the mass is not the only factor here)

Observational properties:

- **distance of objects** using the relationship between **brightness** and **luminosity**.
(**standard candles** = Cepheid or variable stars, HR diagram for cluster, supernova type IA
Or speed of rotation of galaxies)

OR Using radar or parallax method if objects not too far
(we use parallax to calibrate the standard candles).

$$B = L / 4 \pi d^2 \quad (\text{inverse square law})$$

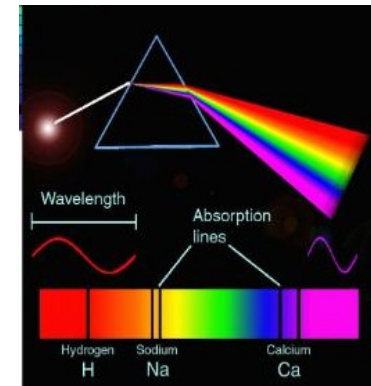
$$p = 1 / d \quad p \text{ in parsecs, } d \text{ in arc seconds}$$



- relative radial speed of stars or galaxies and rotational speed

From Doppler shift of spectra . From relative speed of galaxies = expansion of Universe.

From relative radial speed of stars we detect exoplanets and use Kepler's law to find masses
(show video spectrum moving)







So they collected:
 → luminosity
 → spectra of stars =
 continuum with absorption lines

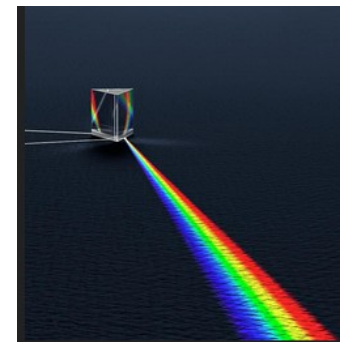
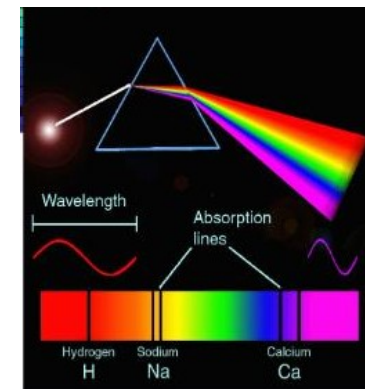
Spectroscopy started
 In the late 19th century

When Astro photography
 Was developed

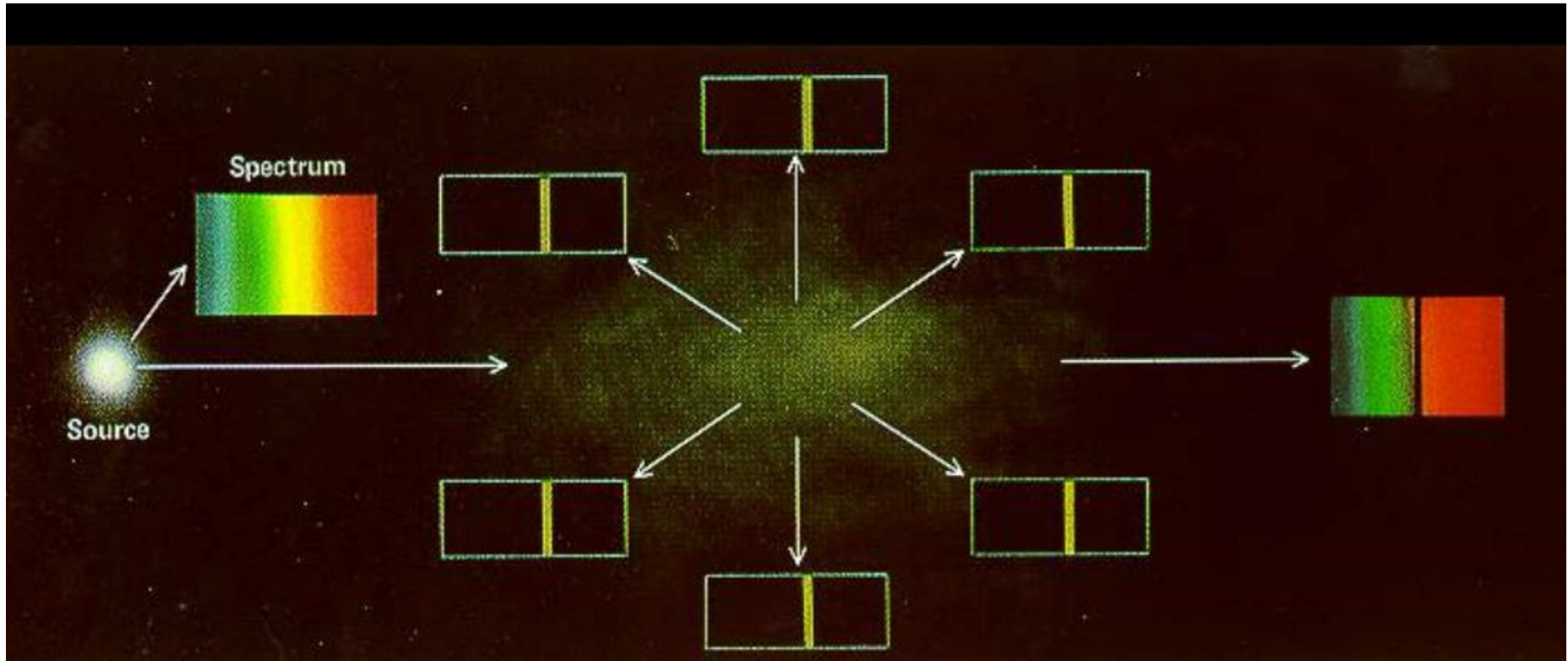
Camera were attached
 To telescope and they
 Increase the time of exposure →
 Brightness, color.

From brightness and distance
 → Luminosity.

Also. A prism was inserted
 Between the film and the camera.
 → stellar spectra

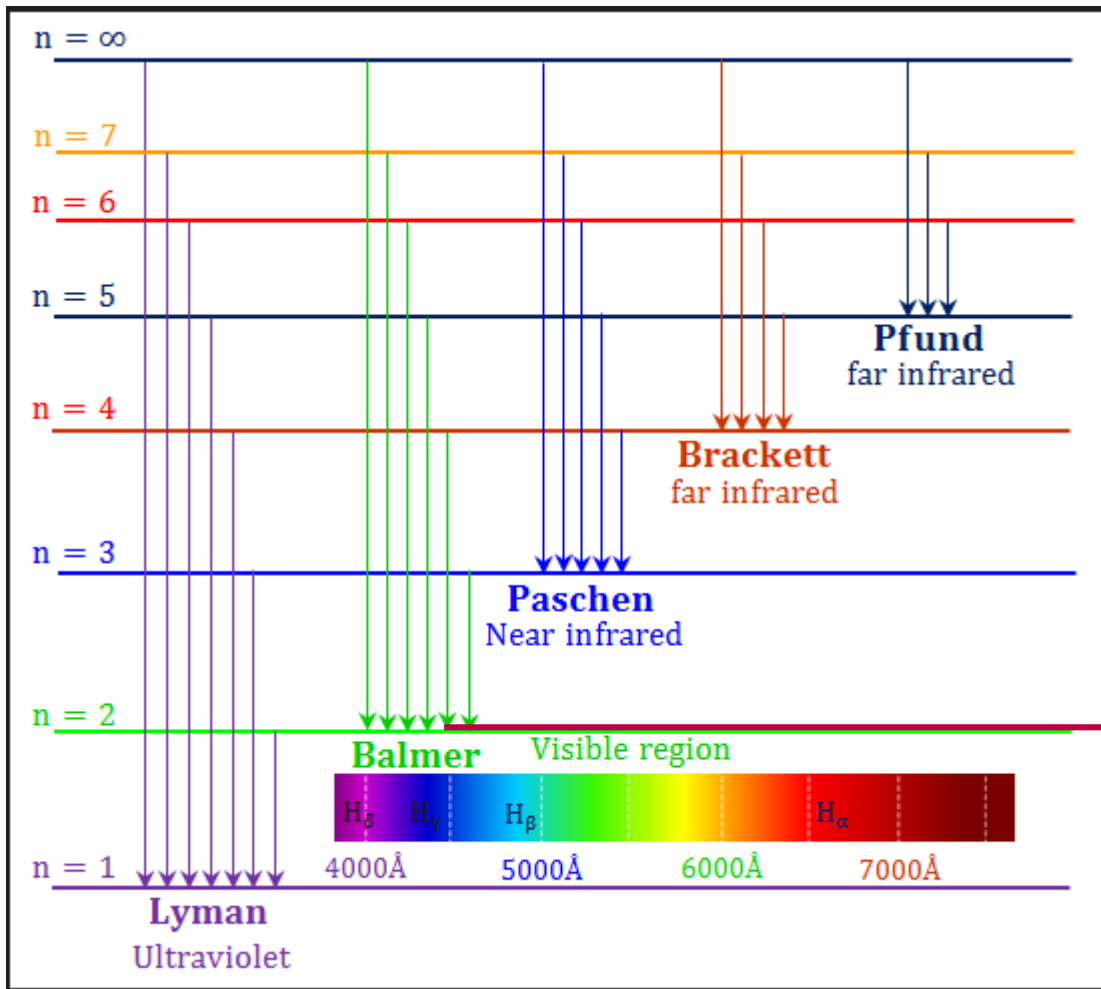


Show video of sodium

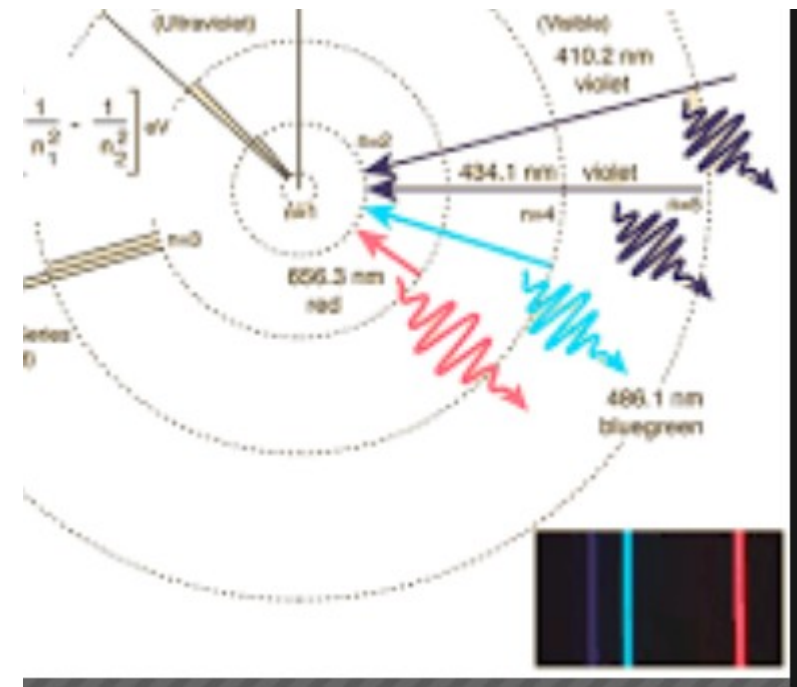


Sodium Absorption Lines:

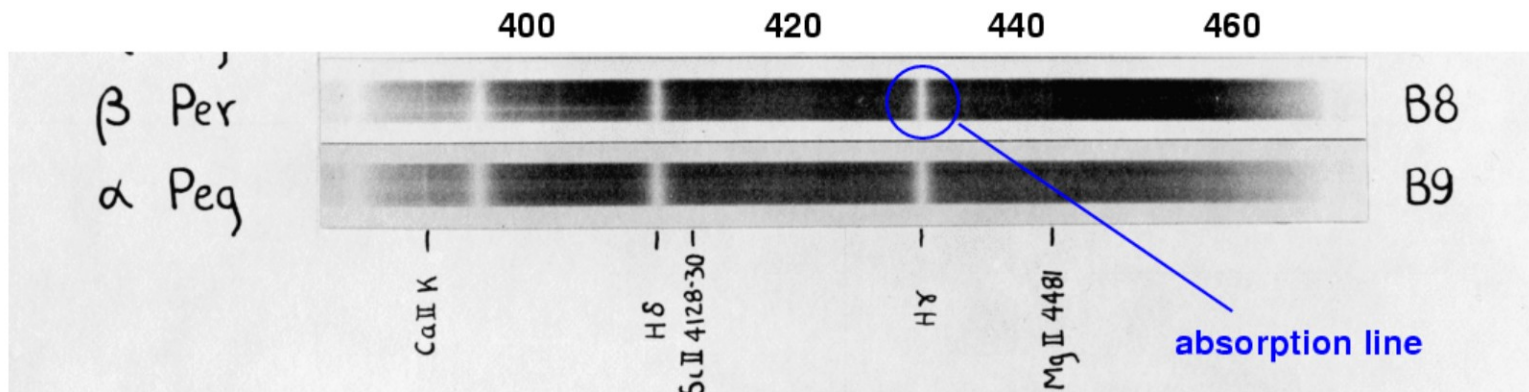
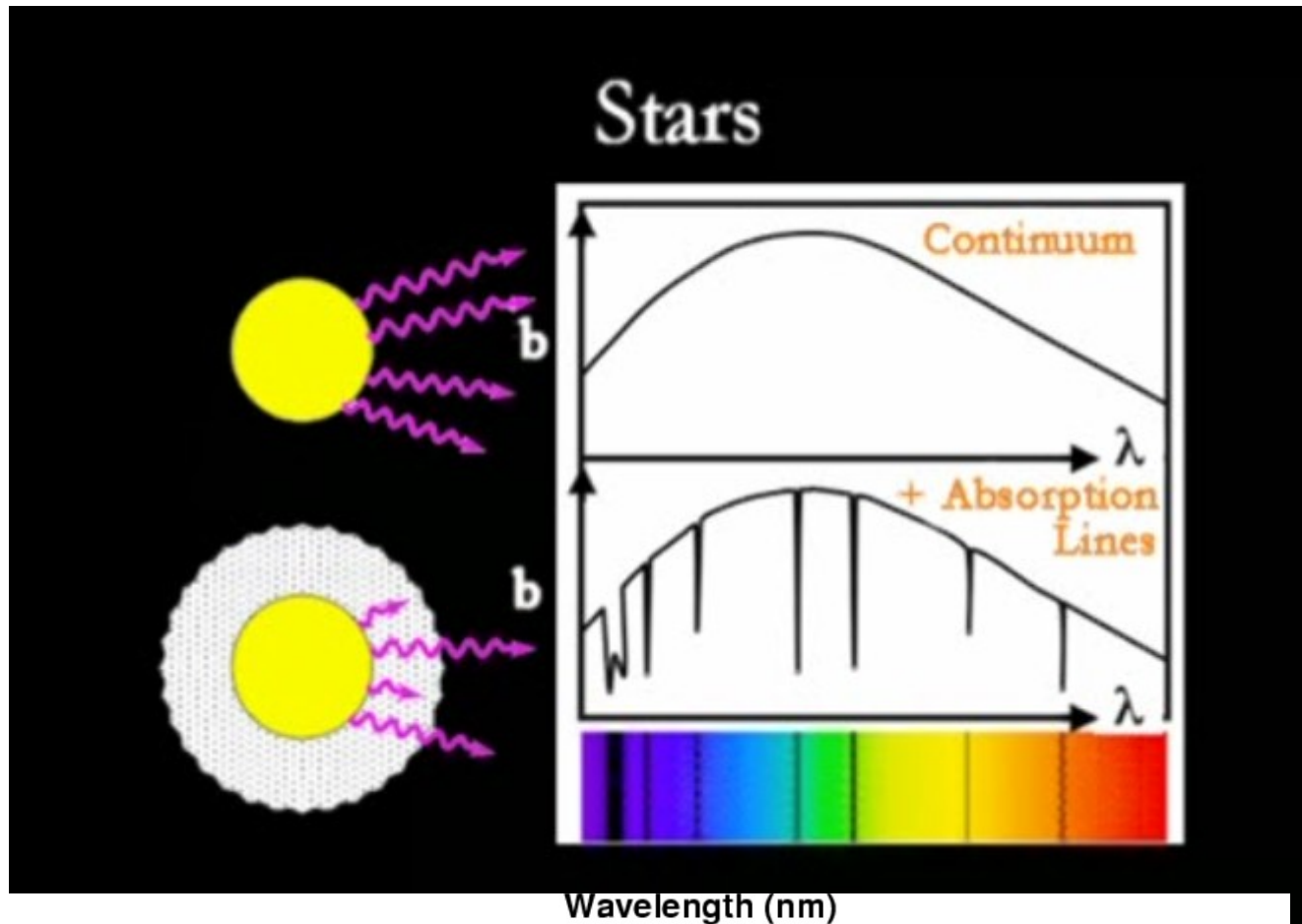
The sodium vapor “subtracts out” the yellow lines from the continuous spectrum emitted by the source



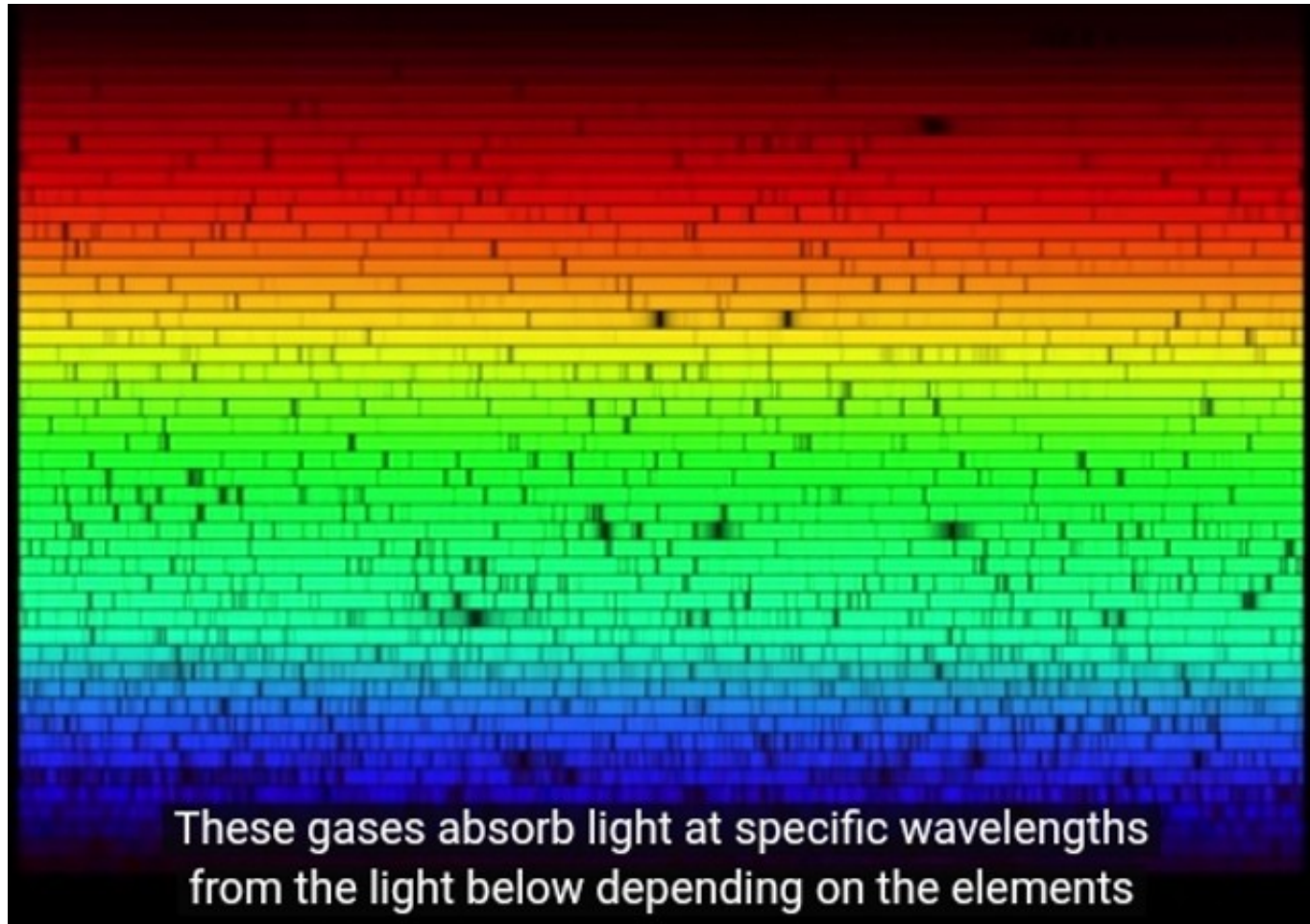
Simulation
Exploration physical



Spectra of stars were collected. A catalog of stars. Using photographic plates
The black lines are like a QR code.
→ which elements are in the atmosphere of the star.



Spectra of stars → All we know about them. Light is the great informant.



2 videos about spectrum of stars.

Pioneers of Stellar Classification

- Annie Jump Cannon and the “calculators” at Harvard laid the foundation of modern stellar classification.



They were called the Harvard computers. They worked at the Harvard College Observatory (HCO) . Their job was to examine photographic plates of spectral lines. This was under the Supervision of Edward Pickering. The group was referred as “ Pickering Harem”.

Their wage were minimum. It was a tedious and boring job. Some went to women colleges

First the stars were classified according to their hydrogen lines.
(thickness) . Those lines were not understood until Physics stepped in.

Annie merged/ dropped the original classes and simplified the classification.

In 1911, she was appointed as the Curator of Observational
Photographs at Harvard's observatory.

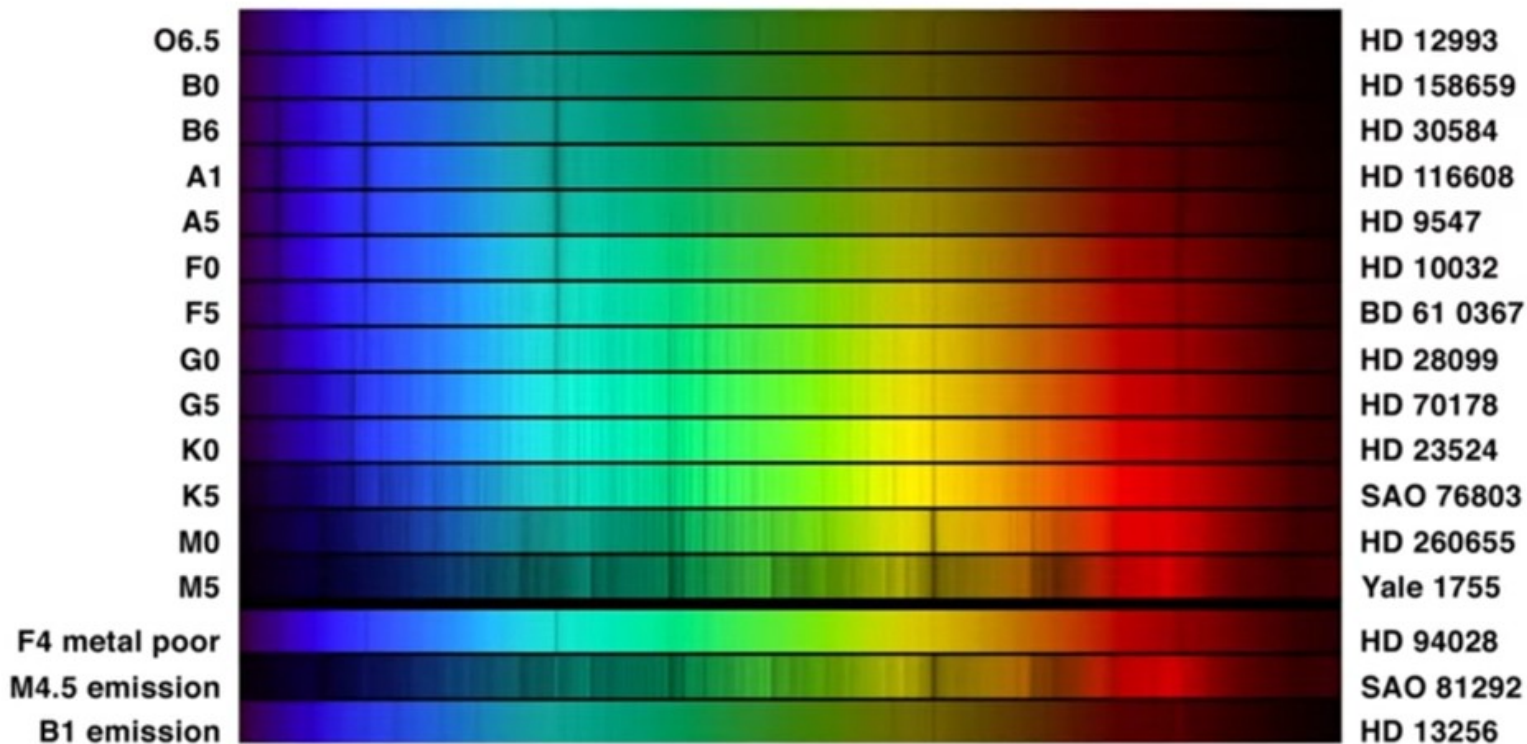
She was known for her speed at classifying stars and
was credited with classifying 5,000 stars per month from 1911 to 1915.

Annie developed a new system still used today.

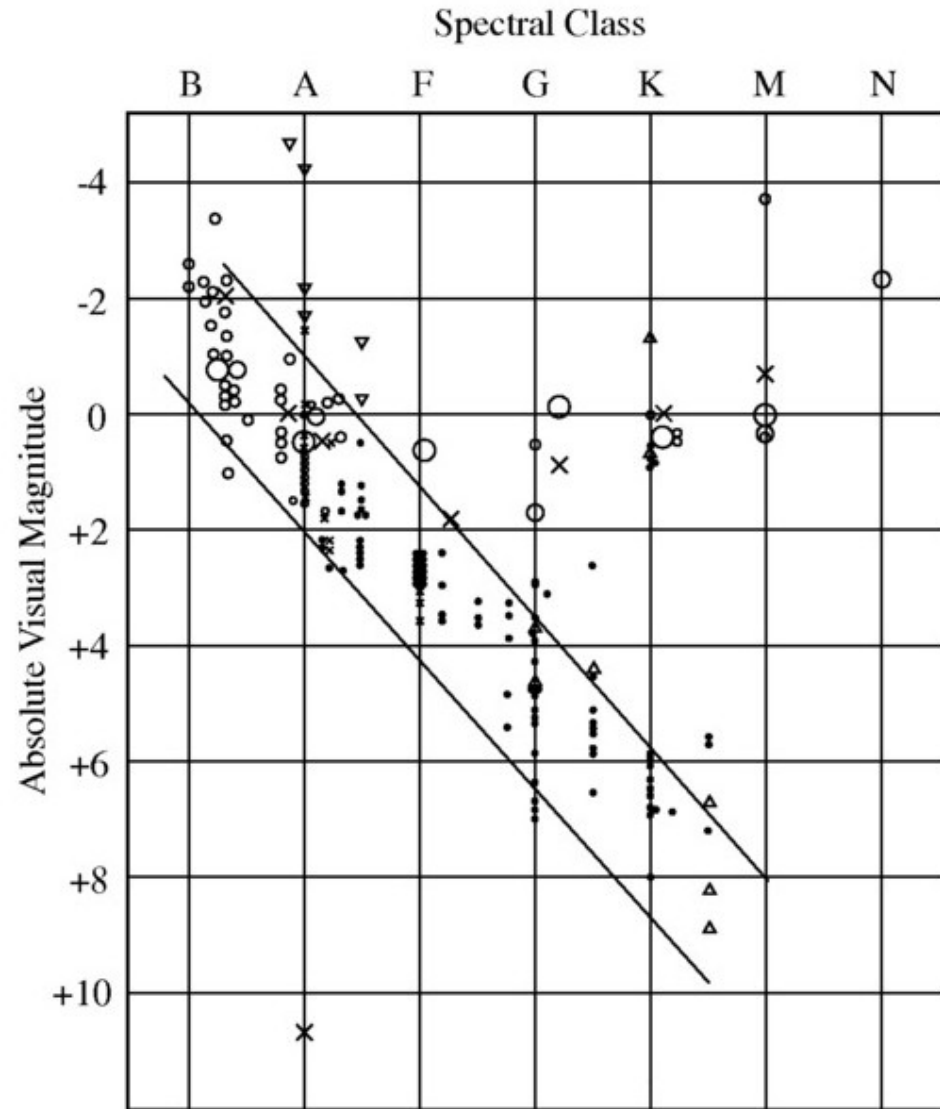
She merged, dropped and classified stars by the many absorption
Lines in their spectra. Still used today. See the cute video about Annie.
She is the one who came with “Oh boy a fine guy/girl kiss me Lovingly “



Annie Jump Cannon
(1863-1941)



HR diagram : One of the most useful and powerful graph in astronomy
Like the periodic table in chemistry.



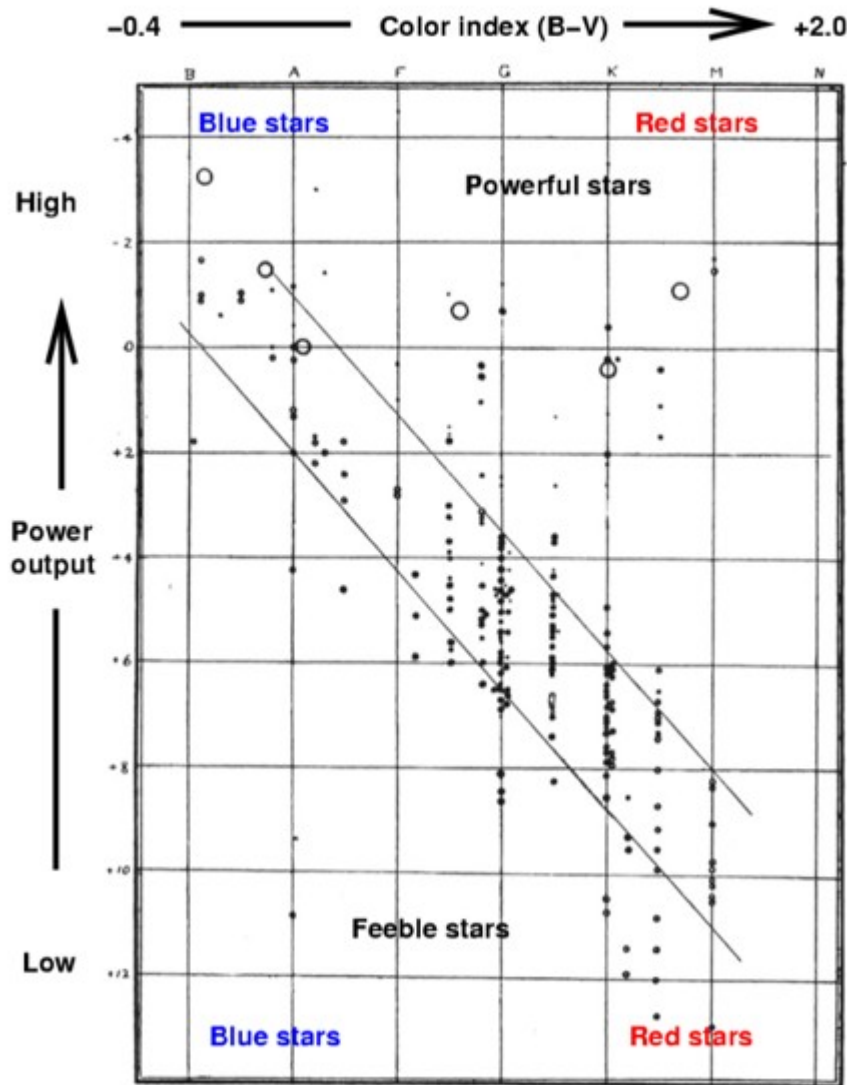
Henry Norris Russell and Ejnar Hertzsprung made independent contributions to the development of the HR diagram

Hertzsprung: **Ejnar Hertzsprung**, a Danish astronomer, is known for plotting the absolute magnitudes of stars against their spectral types (which were related to temperature) in 1911. He found that there was a correlation between a star's spectral type, and its intrinsic brightness (luminosity). However, he didn't directly use colors in his early work.



They use the spectra of stars cataloged by the computer. They knew the distance of stars like the Pleiades (computers) so They were able to find the luminosity (absolute magnitude) using Long exposure photography

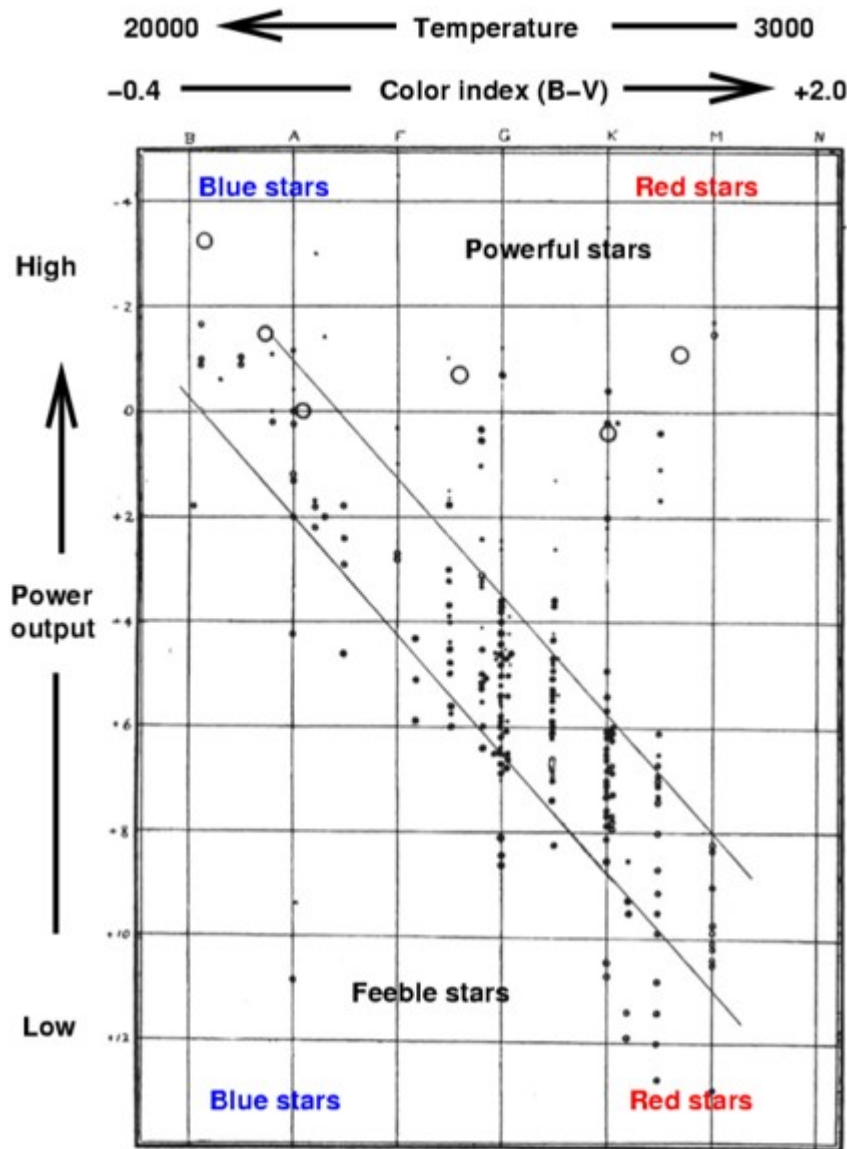
How did they get the luminosity? Using Astro photography and long exposure.



Russell: **Henry Norris Russell**, an American astronomer, developed a similar diagram around the same time, although he used colors rather than spectral types. In 1913, Russell plotted the colors (expressed as color index or B-V) of stars against their luminosity. He recognized that stars with similar colors were clustered along certain sequences or "tracks" in the diagram.



How did they get the luminosity? Using Astro photography and long exposure.

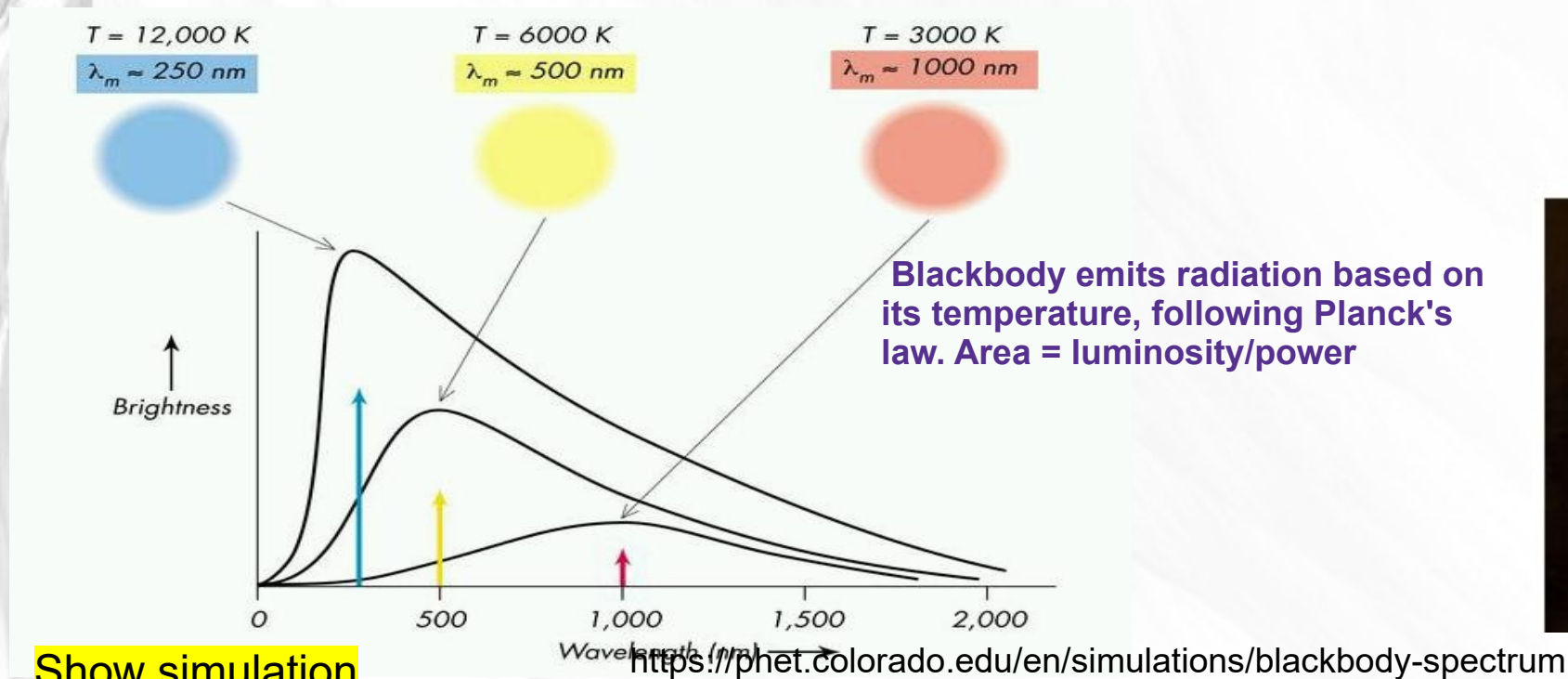


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Planck solved one piece of the puzzle. Using spectroscopy and finding the peak wavelength → you find the temperature. Each spectral class is associated with a range of temperature. See picture in folder .

**The surface temperature of a star only depends on the peak Wavelength.
The energy output depends on the area under the Curve.**



Show simulation

A white star emit the same amount of blue yellow ..and red photons.
The sun glows white hot. As an iron rod gets hotter :
From red, to orange, yellow, to white ..

BLUE STARS Are HOT !!! and RED STARS are COLD (relatively)

Red stars are cool, with temperatures of around 3,000 degrees Kelvin (K), while blue stars are hotter and can have temperatures over 30,000 degrees K. Our own lovely yellow Sun's temperature is a comforting 6,000 degrees C (10,000 F).

Blackbody Radiation

Sun

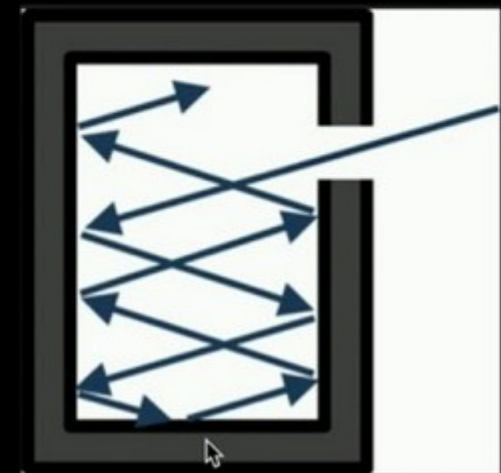


Mustafar



- Two best examples of **blackbody radiation** (BBR) are the sun and hot stove, both of which emit red light due to their temperature.
- When heated, the molecules comprising a perfect blackbody vibrate and emit light of the same wavelength as their vibration.
- Even the fictitious planet, **Mustafar**, emits BBR due to its immense quantity of lava.

Hot Stove



Thermal emitters (or black body) glow mostly at a given wavelength given its temperature

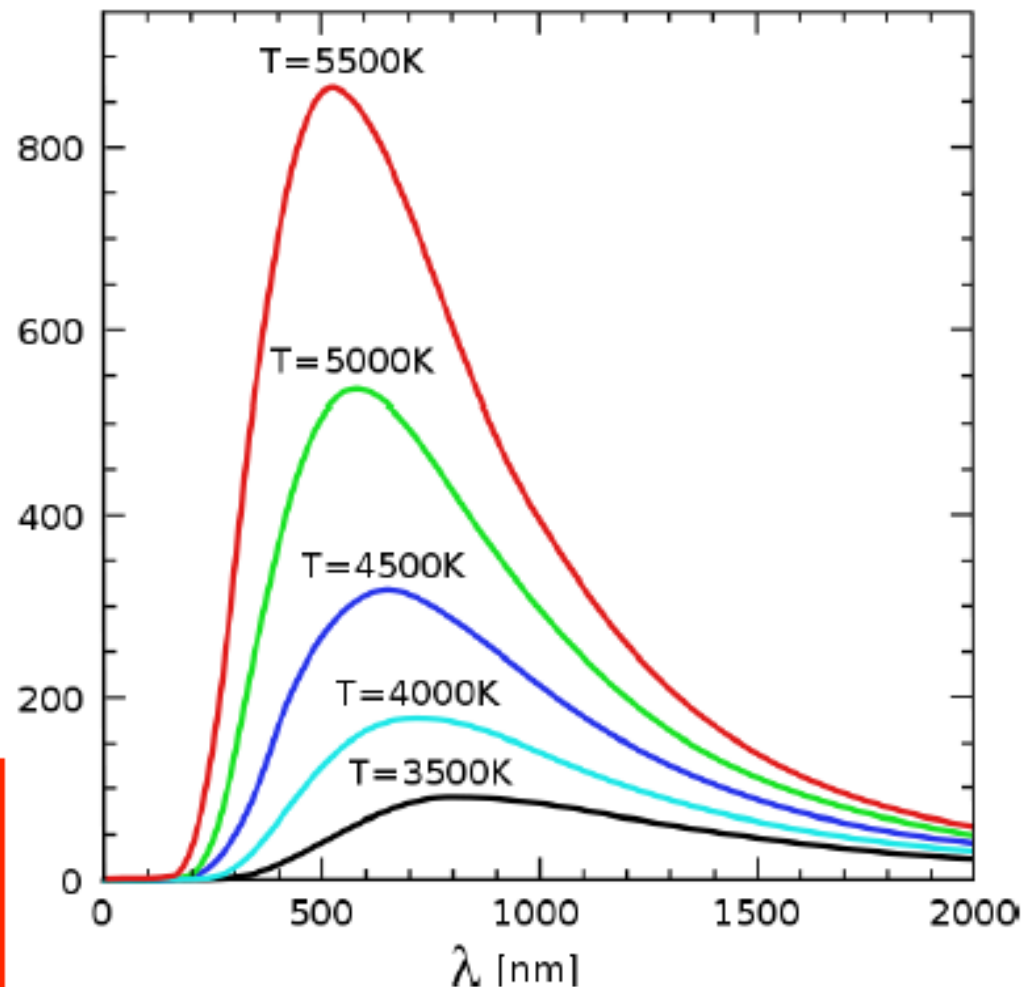
Hotter objects emit
shorter λ

Cooler objects emit
longer λ

$T(K)$	$\lambda_{\text{peak}} (\text{\AA})$
3500	8300
4000	7300
4500	6400
5000	5800
5500	5300

Wien's Law

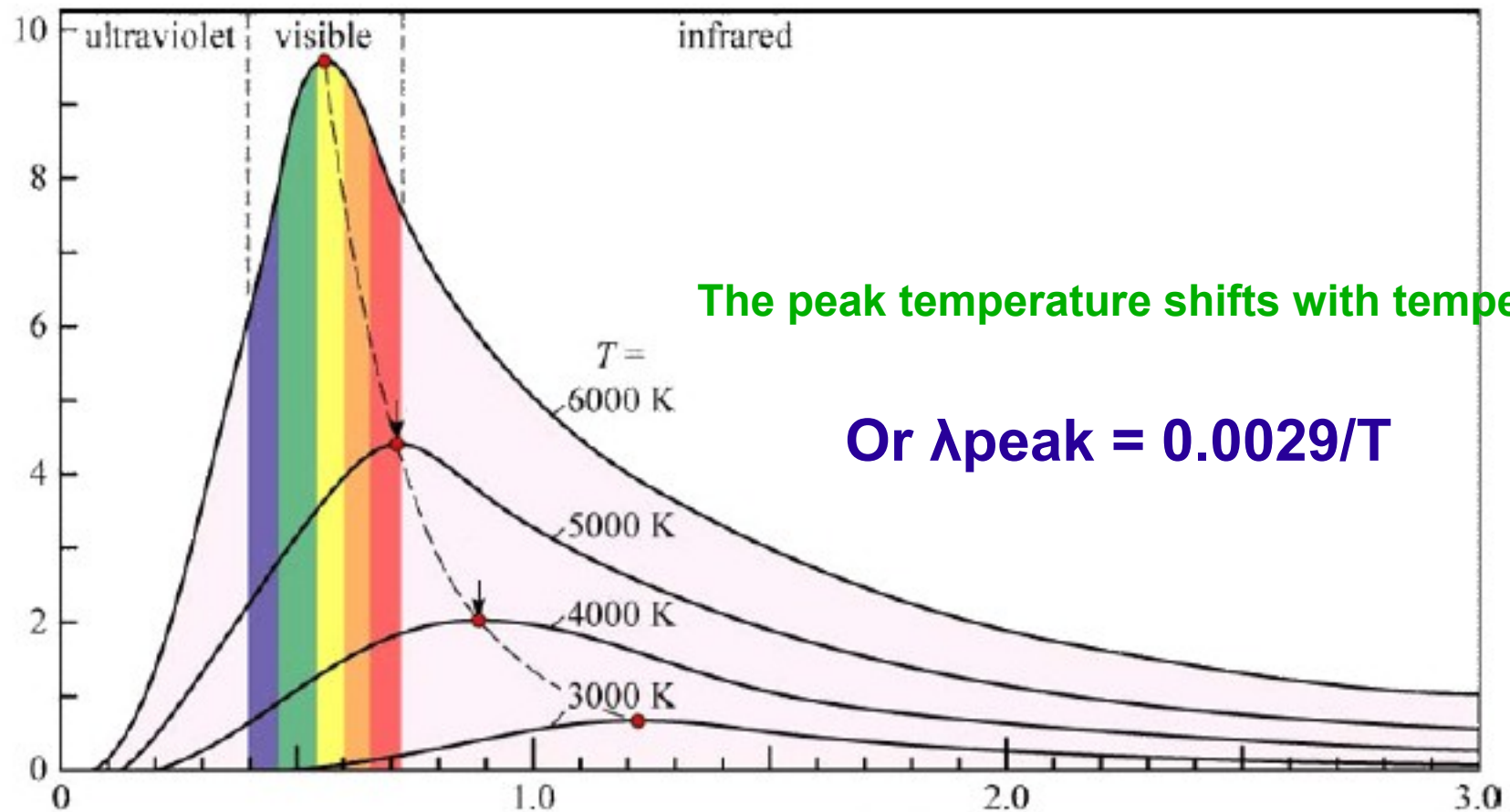
$$\lambda_{\text{peak}} T = \text{constant} \\ \approx 2.9 \times 10^7 \text{ \AA K}$$



$$\text{Or } \lambda_{\text{peak}} = 0.0029/T \text{ (in m)}$$

WE GLOW IN INFRA RED !

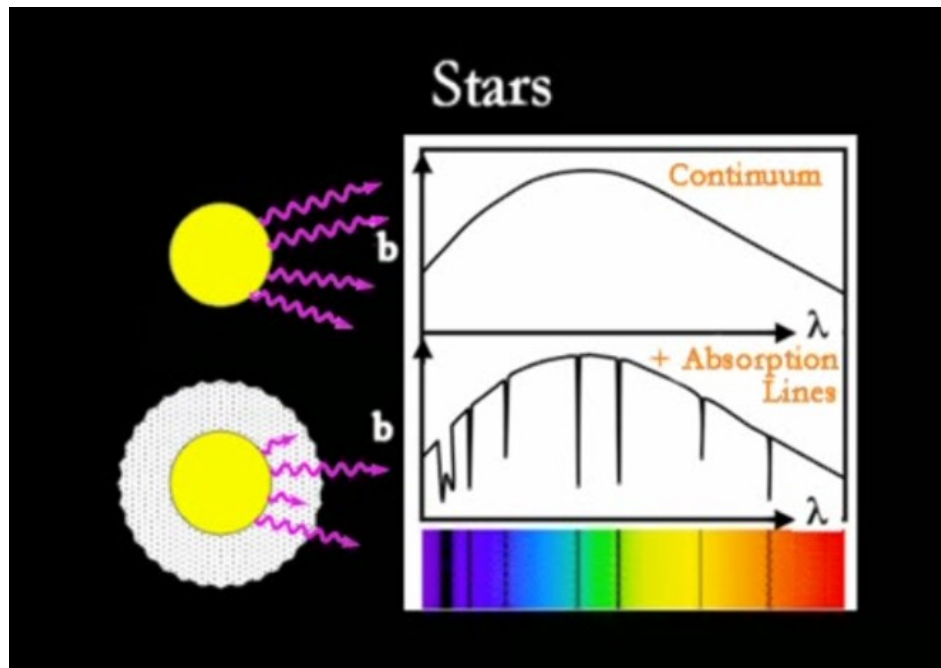
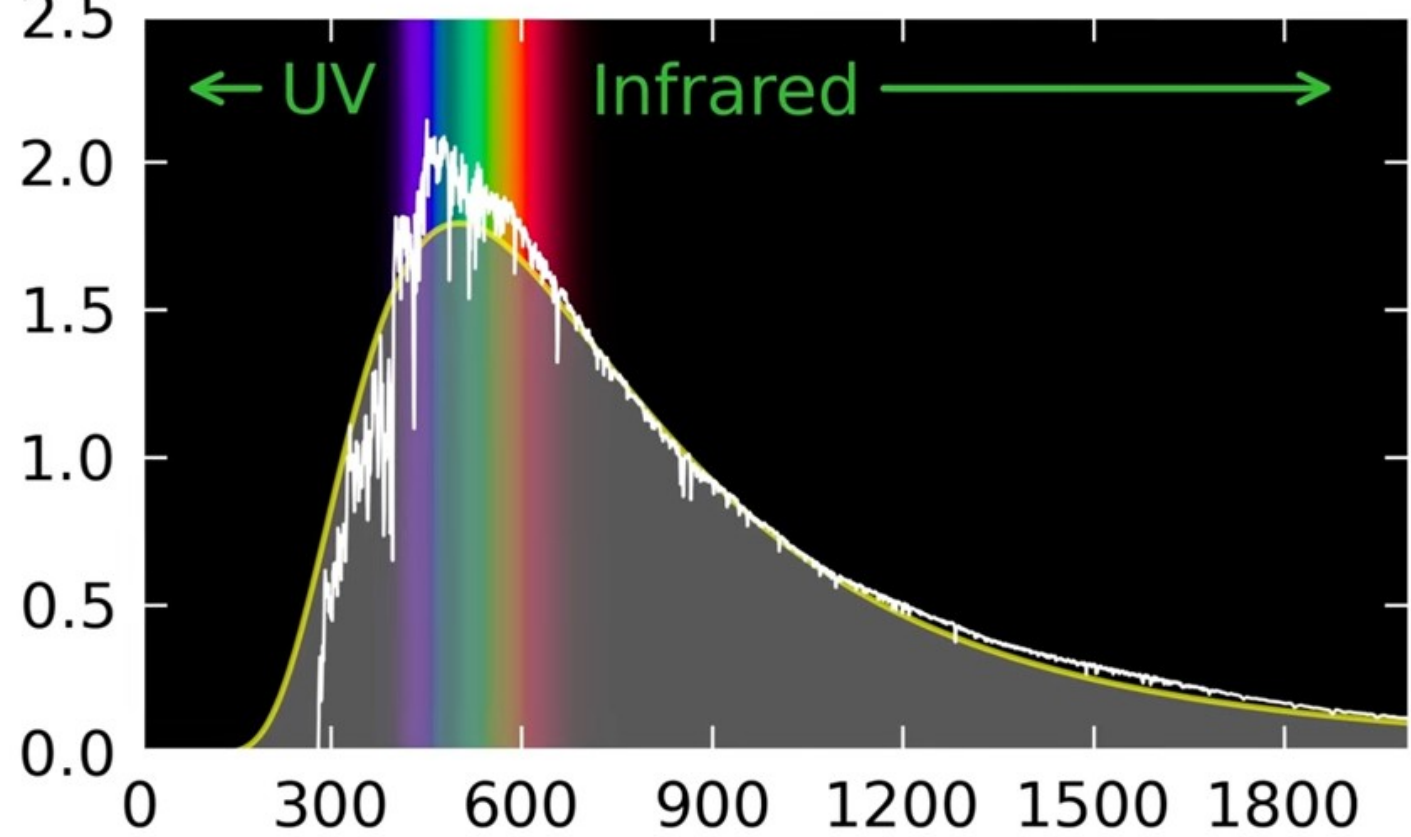
Different Temperature Black Bodies



The peak temperature shifts with temperature

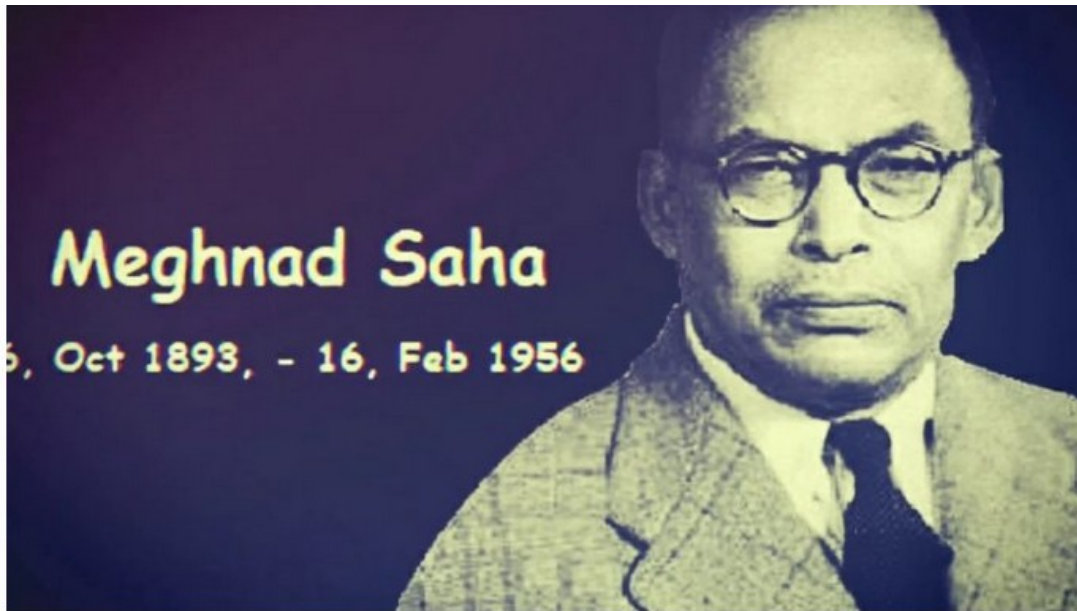
$$\text{Or } \lambda_{\text{peak}} = 0.0029/T$$

A 5000 K star peaks in the red, while a 6000 K star peaks in the yellow. Even hotter stars are blue. **Is this backwards to our intuition?!**



simulation

Last piece of the puzzle: Meghnad Saha
Explained the QR codes as a function of
temperature.



What Stars Are Made Of

The Life of Cecilia Payne-Gaposchkin

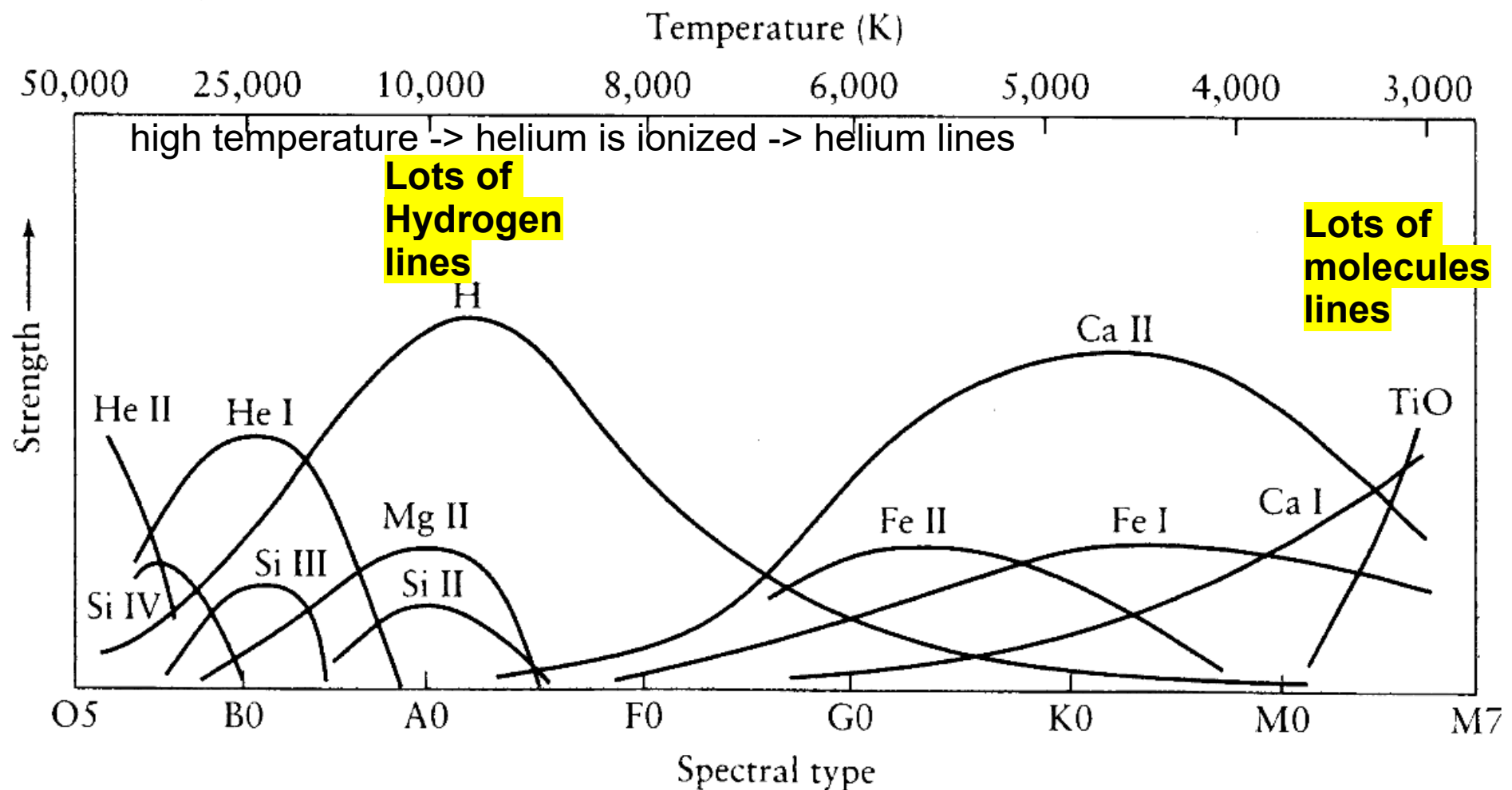
480 x 725

Meghnad Saha. His work allowed astronomers to accurately relate the spectral classes of stars to their actual temperatures. He explained how atoms give up light at different temperature. See next slide.

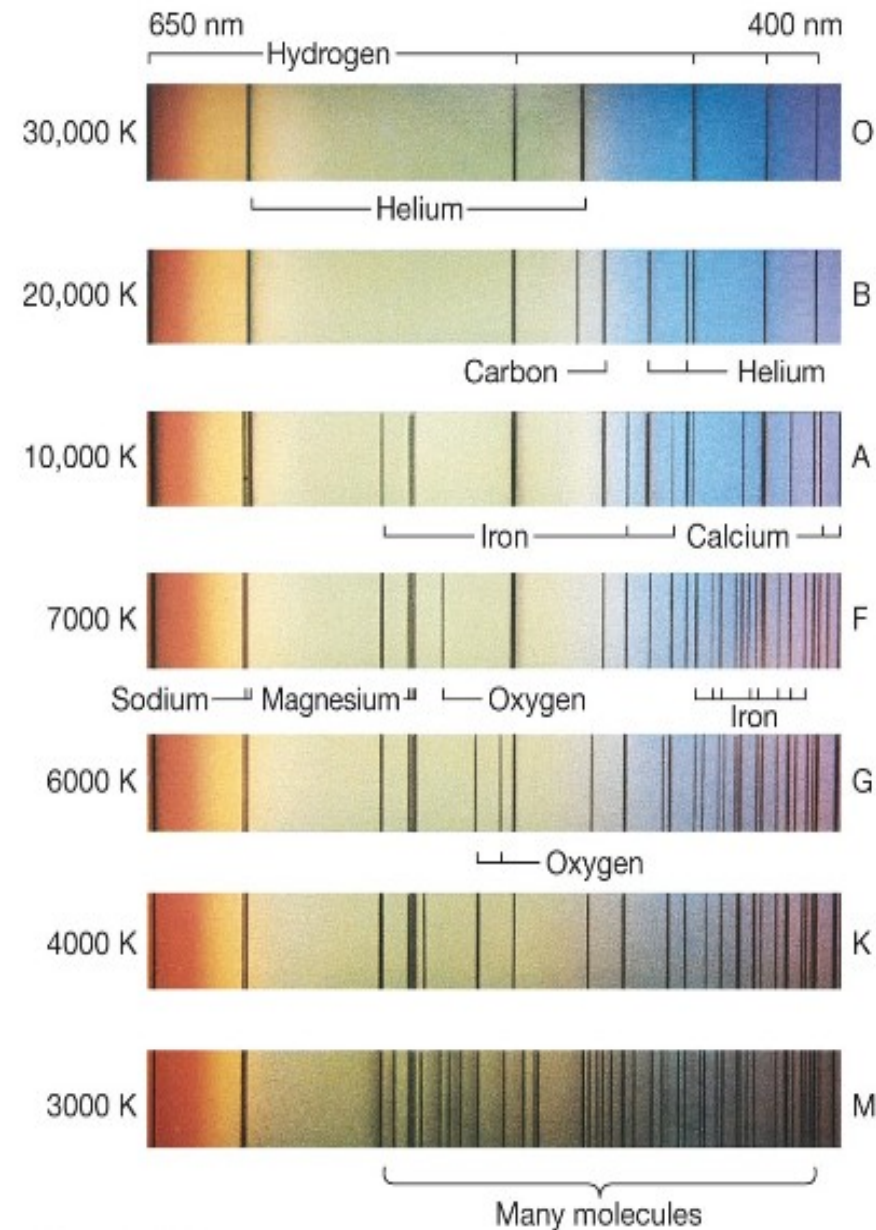
Cecilia Payne 2 decades solidified the concept. British-born American astronomer who discovered that stars are made mainly of hydrogen and helium and established that stars could be classified according to their temperatures. Payne entered the University of Cambridge in 1919.

Payne concluded that, unlike on Earth, hydrogen and helium are the dominant elements of the Sun and stars. Henry Norris Russell strongly opposed this conclusion and convinced her to omit it from her thesis

in 1925 she became the first person to earn a PhD in astronomy from Radcliffe College of Harvard University.



Different atoms and ions leave their fingerprints on starlight at different temperatures. Conditions must be fine-tuned to bring out a given chemical 'fingerprint.' Hydrogen, for instance, only shows up strongly in spectra near type A — even though it makes up the bulk of almost all stars. And hundreds of lines from 'metals' (elements heavier than helium) dominate the spectra of cooler K-type stars, even though metals make up just a tiny fraction of those stars' masses.

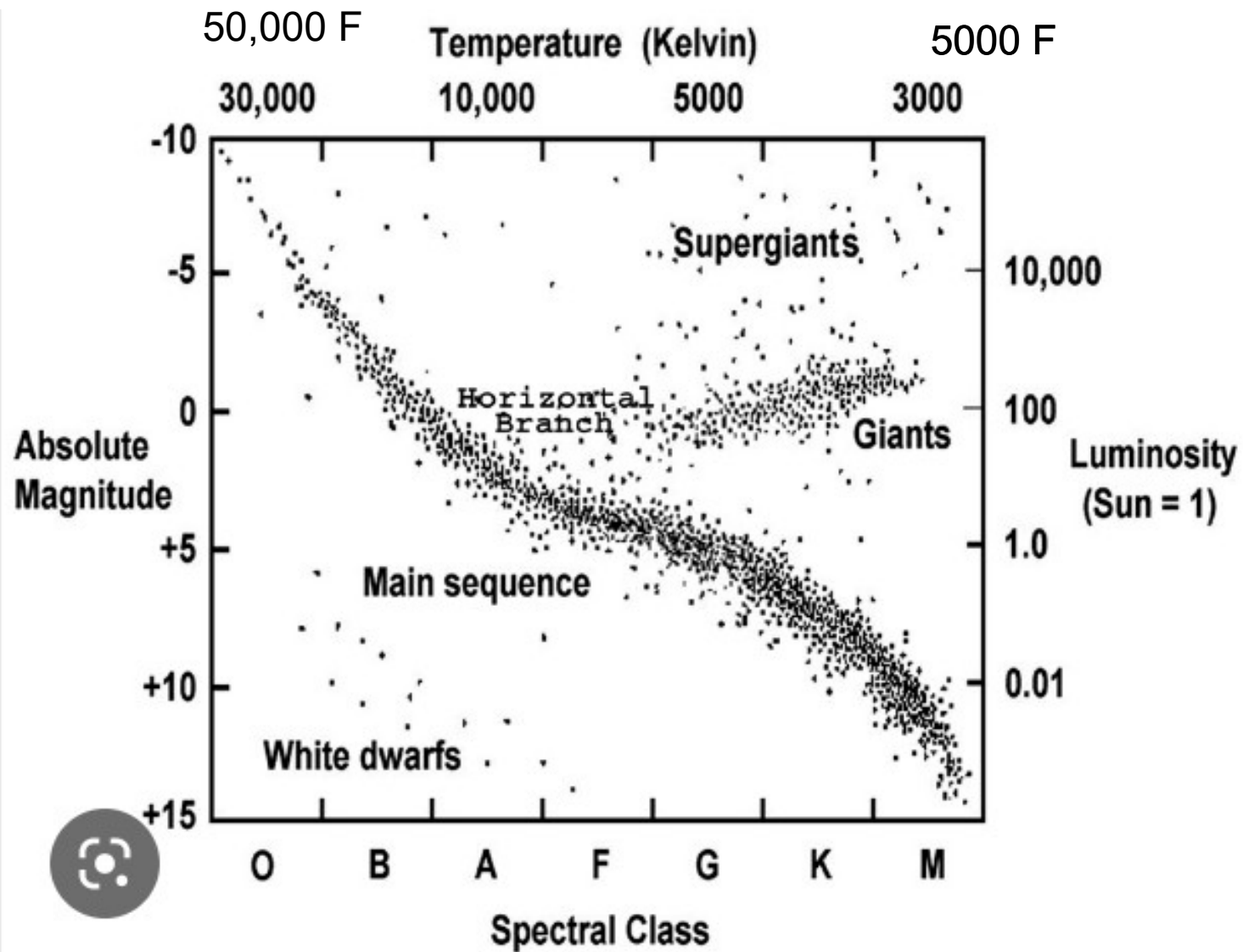


Spectral Type	Color	Temperature (K)*	Spectral Features
O		28,000-50,000	Ionized helium, especially helium
B		10,000-28,000	Helium, some hydrogen
A		7,500-10,000	Strong hydrogen, some ionized metals**
F		6,000-7,500	Hydrogen and ionized metals such as calcium and iron
G		5,000-6,000	Both metals and ionized metals, especially ionized calcium
K		3,500-5,000	Metals
M		2,500-3,500	Strong titanium oxide and some calcium

* To convert approximately to Fahrenheit, multiply by 9/5.
 ** Astronomers regard elements heavier than helium as metals.

NOW We don't need to use Wien's law (peak Wavelength) to find the temperature → We use the spectral class (spectroscopy)

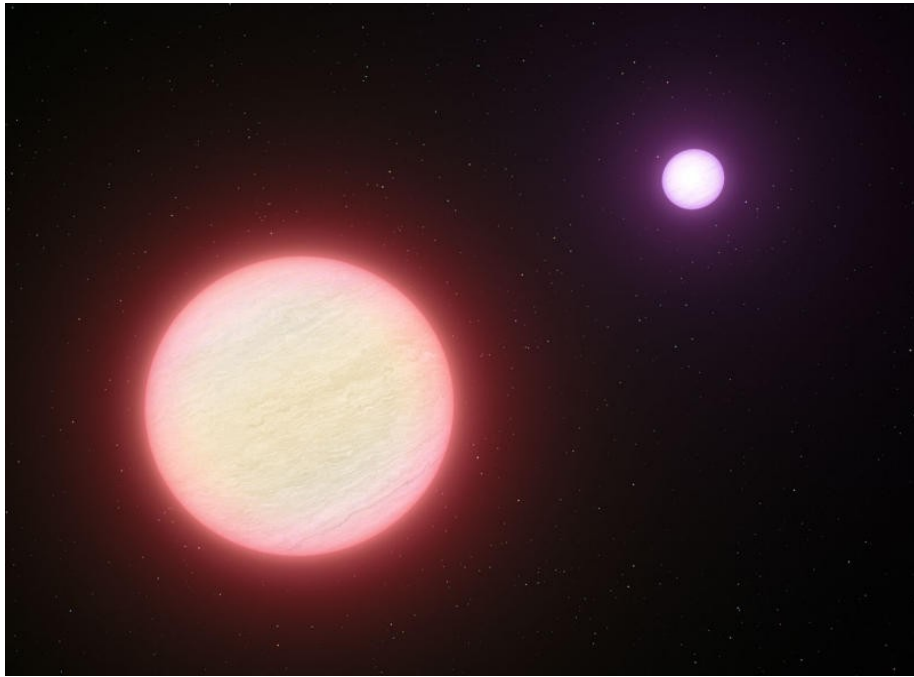
Video: classification (short) and Annie



Our sun is about 6000K = 10,000 F

**Stars come in a wide range of temperatures.
The hottest stars in the sky have temperatures
in excess of 40,000K (70,000F).**

**The coolest stars have temperatures of the
order of 1000-1500K (1300F-2200F)**



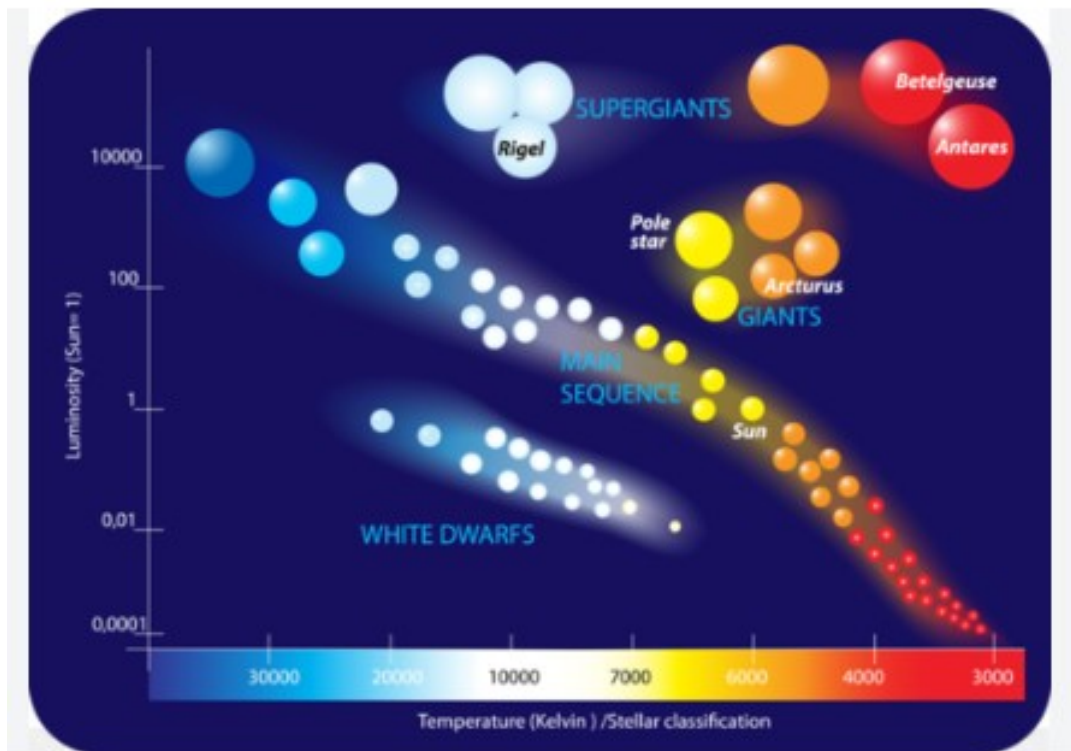
Observations with Very Large Telescope (chile) and two other telescopes have shown that this pair is the coolest pair of brown dwarfs found so far. The colder of the two components (shown in the background) is a candidate for the brown dwarf with the lowest temperature ever found — the surface temperature is similar to that of a cup of freshly made tea. The two components are both about the same size as the planet Jupiter.



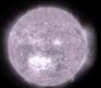
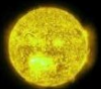
Astronomers using the Very Large Telescope have identified the hottest and most massive contact binary ever seen. This double star system – VFTS 352 – is located 160,000 light-years away (72,032 F)

northern Chile

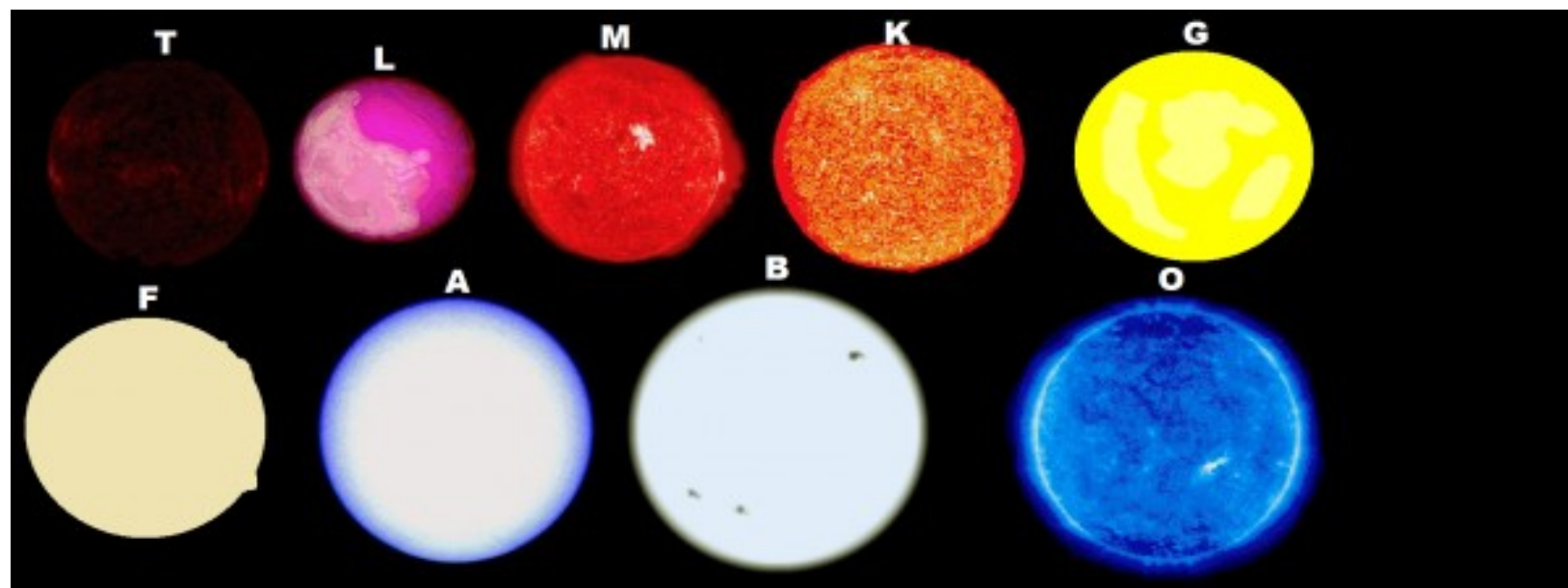
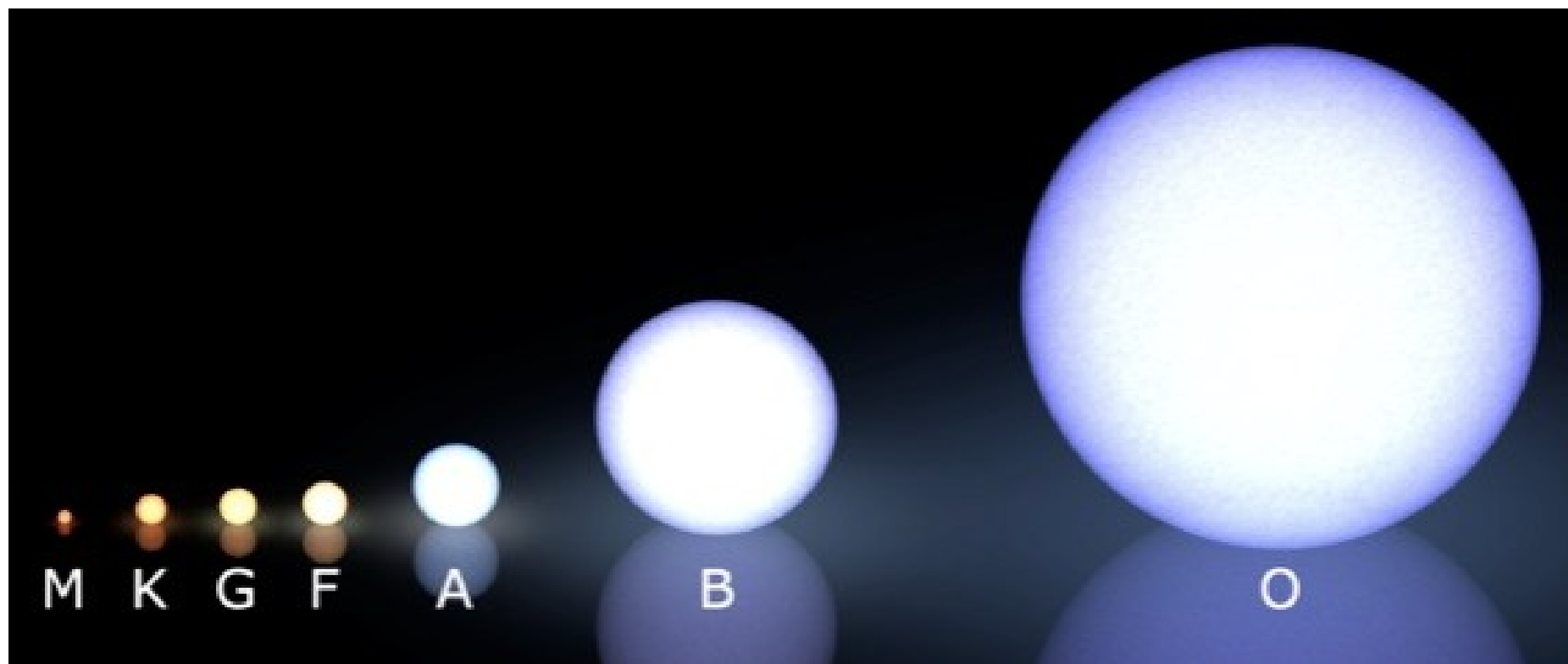
Paranal Observatory is an astronomical observatory operated by the European Southern Observatory (ESO). It is located in the Atacama Desert of northern Chile on Cerro Paranal at 2,635 m (8,645 ft) altitude, 120 km (70 mi) south of Antofagasta.



<https://astrobackyard.com/types-of-stars/>
watch video

COLOR	EXAMPLE	SURFACE TEMP IN CELSIUS
	SPICA	28,000 - 11,000
	VEGA	11,000 - 7,500
	THE SUN	6,000 - 5,000
	ARCTURUS	5,000 - 3,600
	ANTARES	3,600 - 2,000

Spectral Class	Color	Temperature
O	blue	28,000 - 50,000
B	blue-white	9900- 28,000
A	white	7400-9900
F	yellow-white	6000-7400
G	yellow	4900-6000
K	orange	3500-4900
M	orange-red	2000-3500



Properties of Stars (don't memorize!)

Type	M (Msun)	T (K)	L (Lsun)	R (Rsun)
O5	40	40,000	5×10^5	18
B0	18	28,000	2×10^4	7.4
B5	6.5	15,000	800	3.8
A0	3.2	9,900	80	2.5
F0	1.7	7,400	6	1.4
G0	1.1	6,000	1.3	1.1
K0	0.8	4,900	0.4	0.8
M0	0.5	3,500	0.03	0.6
M5	0.2	2,800	0.008	0.3

Life expectancy
Shortest life expectancy is 2 million years
20 million years
Hundreds of millions
Billions
Tens of billions
Trillions of years

The Sun is a G2 star ($T = 5800$ K). By far, the majority of stars in our Galaxy are K and especially M stars (average mass 0.5 Msun). Still, the Sun is a pretty average star.

Can you see any trend ?